

PHYS 230 - Electricity and Magnetism

Lecture 19

Instructor: Saeed Rastgoo



**UNIVERSITY
OF ALBERTA**

Outline

4 Capacitance and Dielectrics

- Energy Stored in a Capacitor (Continued)
- Dielectrics

All figures in these slides are from one of the following sources, unless cited otherwise:

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Exercise I

<https://epoll.srv.ualberta.ca> – Code: MBD

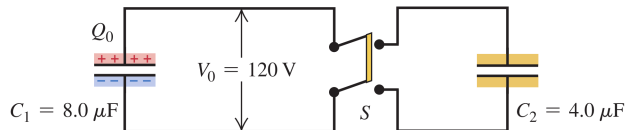
We connect a capacitor $C_1 = 8\ \mu\text{F}$ to a power supply, charge it to a potential difference $V_0 = 120\ \text{V}$, and disconnect the power supply. At this point switch S is open.

(a) What is the charge Q_0 on C_1 ?

(b) What is the energy stored in C_1 ?

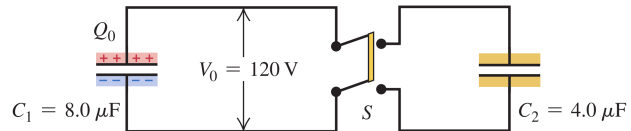
(c) Capacitor $C_2 = 4\ \mu\text{F}$ is initially uncharged. We close switch S . After charge no longer flows, what is the potential difference across each capacitor, and what is the charge on each capacitor?

(d) What is the final energy of the system?



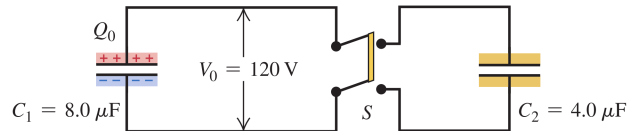
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Exercise 2

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- (a)** What is the magnitude of the electric field required to store 1.0 J of electric potential energy in a volume of 1.0 m^3 in vacuum?
- (b)** If the field magnitude is 10 times larger than that, how much energy is stored per cubic meter?

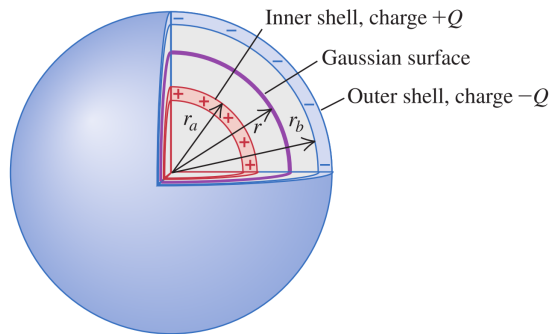
Exercise 3

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A spherical capacitor has charges $+Q$ and $-Q$ on its inner and outer conductors. Find the electric potential energy stored in the capacitor

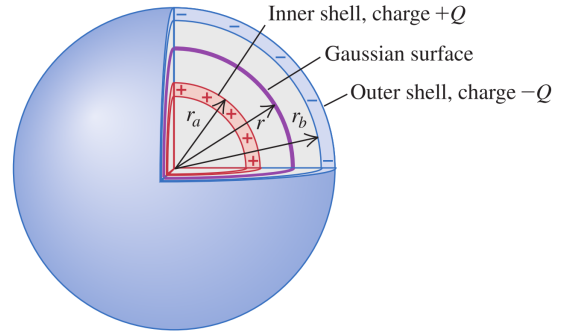
(a) by using the capacitance formula we found before $C = 4\pi\epsilon_0 \frac{r_a r_b}{r_b - r_a}$

(b) by integrating the electric-field energy density u



Exercise 3

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Dielectrics and Capacitors

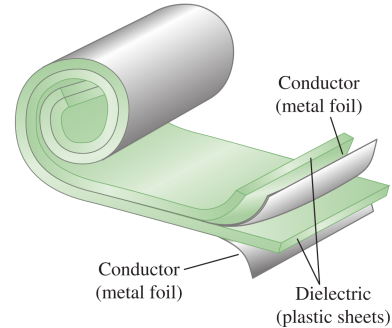
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Dielectrics and Capacitors

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They are places between the conducting plates of capacitors for three reasons:

Figure 24.13 A common type of capacitor uses dielectric sheets to separate the conductors.



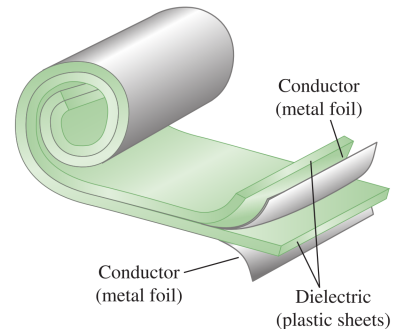
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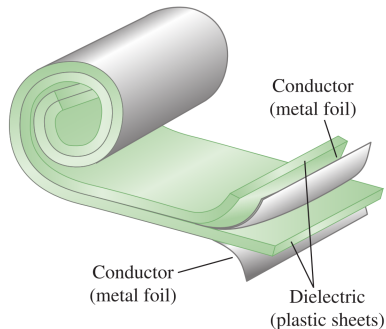
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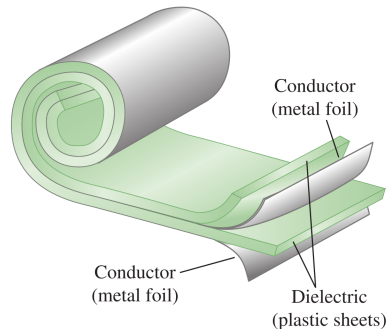
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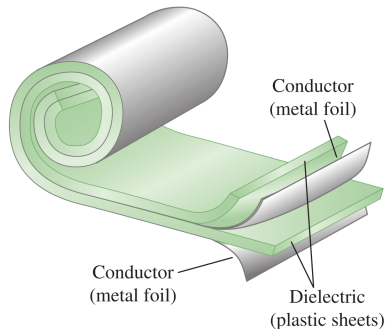
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 - 2.2 Most non-air dielectrics polarize and become conductors at potentials V_d higher than the potential V_{air} that breaks down air
$$V_d^{(\text{max})} > V_{\text{air}}^{(\text{max})}$$

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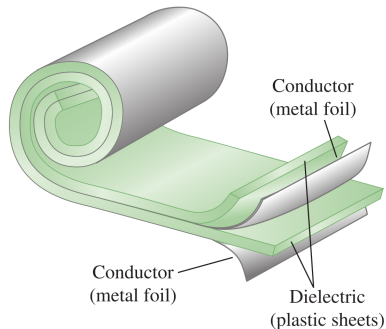
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3. With the same potential (not changing V) if we replace air between the plates with a dielectric, we get **a higher capacitance C** (see next slide)

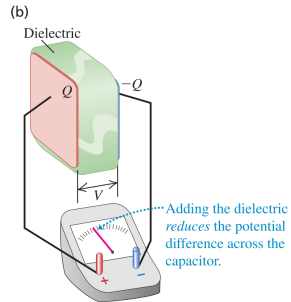
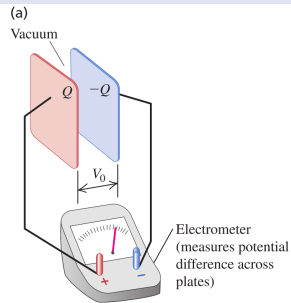
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Capacitance With a Dielectric

A capacitor with air between the plates and charge Q on the plates shows a potential difference V_0 across, so

$$C_0 = \frac{Q}{V_0}$$



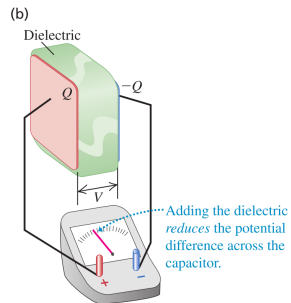
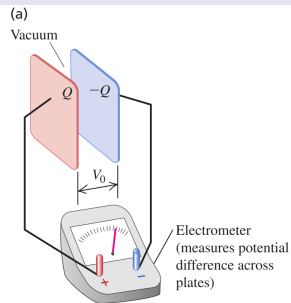
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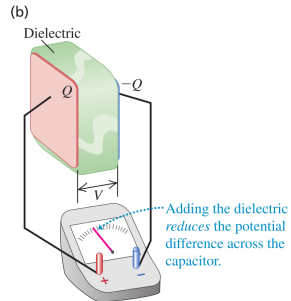
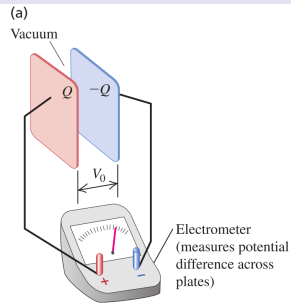
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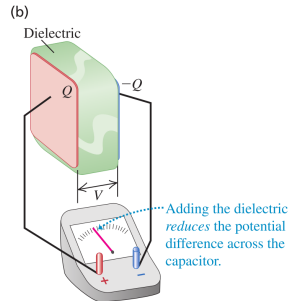
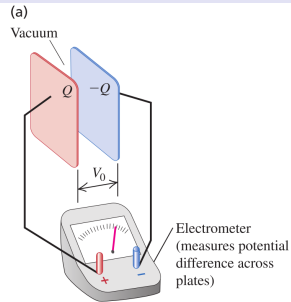
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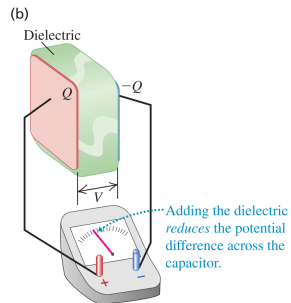
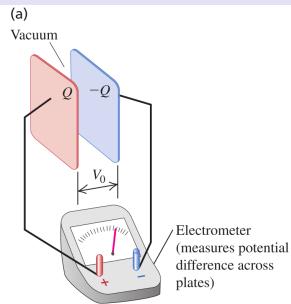
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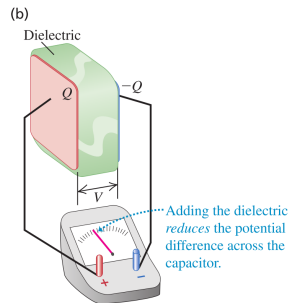
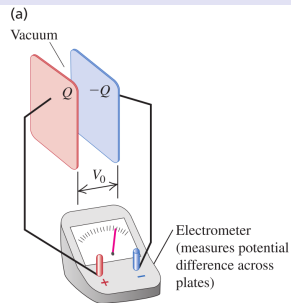
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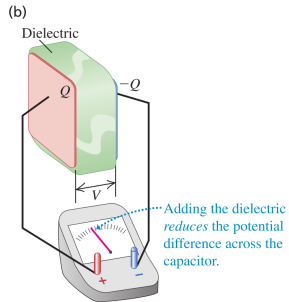
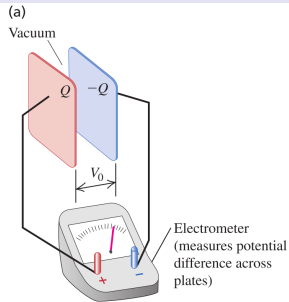
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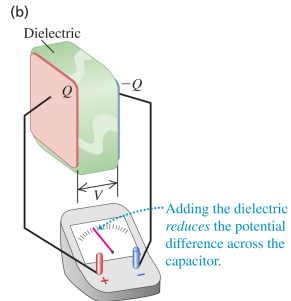
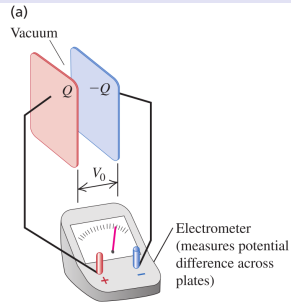
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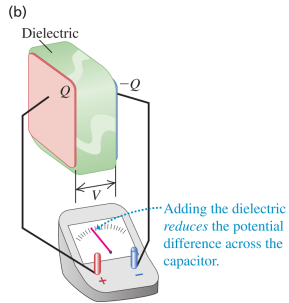
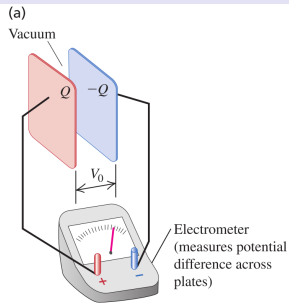
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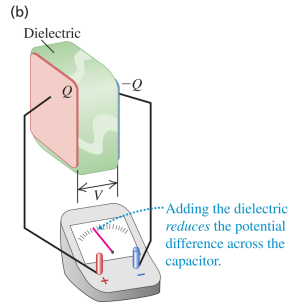
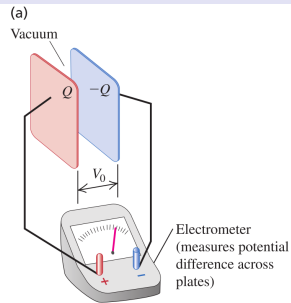
New capacitance C with dielectric is larger than C_0 with air



Capacitance With a Dielectric: Dielectric Constant

When the space between plates is completely filled by a dielectric, the ratio of $\frac{C}{C_0}$ is called the **dielectric constant** of the material, K :

$$K = \frac{C}{C_0}$$



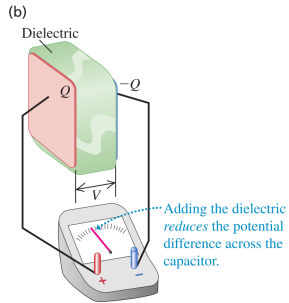
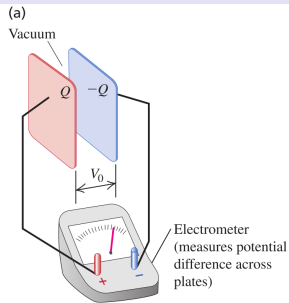
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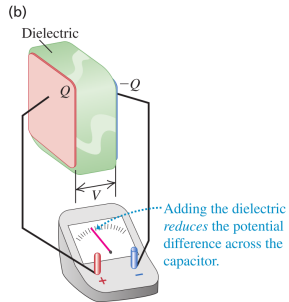
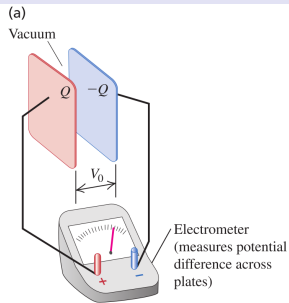
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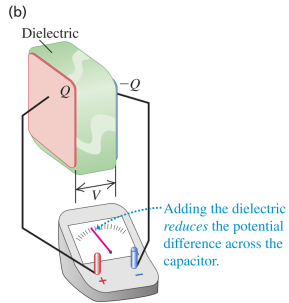
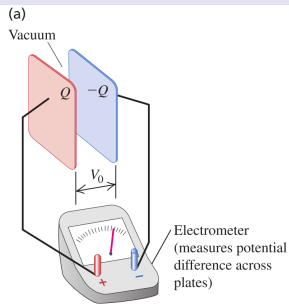
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Note 3: K is a **dimensionless number** related to the **properties of the material**; ϵ_0 is independent of material and a universal constant.



Dielectric Constants

TABLE 24.1 Values of Dielectric Constant K at 20°C

Material	K	Material	K
Vacuum	1	Polyvinyl chloride	3.18
Air (1 atm)	1.00059	Plexiglas [®]	3.40
Air (100 atm)	1.0548	Glass	5–10
Teflon [®]	2.1	Neoprene	6.70
Polyethylene	2.25	Germanium	16
Benzene	2.28	Glycerin	42.5
Mica	3–6	Water	80.4
Mylar [®]	3.1	Strontium titanate	310